IMPLEMENTATION PLAN · v1.1 · 14 OCT 2026

KJS India Digital Platform

Website with Koko AI · Employee portal · Admin console

Auto task assign · Order Boards · Mail & WhatsApp alerts · Superset analytics

How I will build it, step by step

Prepared for: Vineet Pandey (developer, working with Claude Code)

Companion to: kjsKoko.html (15 wireframe screens)

Base documents: MVP Engineering Roadmap v1.0, Hi-fi Wireframes v1.0

Plan window: Wed 14 Oct 2026 to Fri 11 Dec 2026 (8 build weeks + setup)

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1 Summary and key decisions

We are building one platform with three faces: a **public website** where visitors browse best sellers, unlock categories by sharing their contact details, and chat with **Koko**; an **employee portal** with attendance, an availability toggle, auto-assigned tasks and personal stats; and an **admin console** that controls everything, including auto task assignment, **Order Boards** with time-tracked steps, deadline alerts by email and WhatsApp, approvals, a full audit log and **Insights & Analytics** in Apache Superset.

Key decisions

Decision	Choice	Why
Stack	React + TypeScript + Vite, FastAPI (Python 3.12), PostgreSQL 16 + pgvector	Same stack as the MVP roadmap. One language per side, and Claude Code knows it well.
Background work	PostgreSQL job queue (FOR UPDATE SKIP LOCKED) + APScheduler	Batch auto-assign, alerts and re-indexing run without Redis/Celery, so there is one less service to manage.
Analytics	Apache Superset (self-hosted, Docker), embedded in the admin console	Heatmaps, pies and bars without hand-coding each chart. Read-only access to the database.
Chatbot (Koko)	Claude Haiku-class model via Anthropic API, RAG over our own content	Low cost, multilingual, answers only from KJS data, with a monthly spend cap.
WhatsApp	WhatsApp Business Platform (Cloud API) via Meta or a BSP (Gupshup / Twilio / Interakt)	The only legal way to send automated business messages. Templates need Meta approval.
Email	Resend or Brevo (transactional)	Free tier is enough. Good delivery rates.
Hosting	Docker Compose on the company server + Cloudflare Tunnel	Same as the roadmap: no open ports, free HTTPS, easy to move to cloud later.
Working calendar	Mon to Sat working, Sunday off, plus a holiday table	Used by attendance, auto-assign and every deadline calculation.
Step tasks	Step task templates (admin-managed, versioned) copied into each board	Tasks for each step change from client to client. A template gives a standard start, and each board can be edited without code.

Timeline impact. This scope is larger than the original 4-week MVP. With one developer and Claude Code, the realistic plan is **8 build weeks (~42 working days, so Week 8 has almost no buffer): go-live Fri 11 Dec 2026**. If the client needs something live sooner, the public website and Koko can soft-launch on Fri 13 Nov (end of Week 4) while the portal features keep shipping to staging. Confirm this with your manager in writing.

2 Scope by module

#	Module	What it does	Priority
A1	Header + offers ticker	Offers scroll left to right, pause on hover, managed by admin (text, link, start/end date, order). KJS INDIA logo links to https://www.kjsindia.com/. Language switcher. Employee Login button at top right.	Must
A2	Best sellers	Middle section: top products (admin-pinned or ranked by enquiries). Open to everyone, no form.	Must
A3	Category gate	Opening a category (Coffee, Tea, Confectionery...) asks for name + phone OR email once. Creates a lead tagged with the category. Visitor is remembered.	Must
A4	Koko AI assistant	Bottom-right on every page. Answers from KJS content in the visitor's language and creates an enquiry with a chat summary.	Must
A5	Languages	react-i18next. Launch with English + Hindi (others added via translation files).	Must (EN+HI)

#	Module	What it does	Priority
B1	Attendance	Check-in / check-out, multiple sessions a day, shifts, weekly view, regularization requests, auto-close for forgotten check-outs.	Must
B2	Availability toggle	ON = receives auto-assigned tasks/leads. OFF = receives nothing new. Every change is logged.	Must
B3	Tasks + notes	Record batches from auto-assign, order board tasks, tasks added by the admin, and **tasks employees add for themselves**. Statuses, due dates, client query outcome (successful / lost). **Notes** on every task (timestamped, visible to the employee and admin).	Must
B4	Dashboard stats	Employee: total, completed, pending, ongoing client queries, successfully closed. Admin: the same, summed over all employees.	Must
C1	Auto task assign (batches)	Toggle + duration (weekly / monthly / until off / custom), Mon-Sat. Records are fetched from the database in **batches of N** (20 / 30 / 50 or any number, set by the admin, with an optional per-employee override). A batch goes to an employee who is Available, has **no open batch**, and is mapped to that category (one or more categories per person). Live preview and batch history.	Must
C2	Order Boards	New order board (name + ID ABC117-xxxx), requirement files, 5 steps (Procurement, Packaging, Production, Placement, Track), time split by % or days, members per step (members can add colleagues), **step task checklists from templates** showing completed tasks, countdowns, history.	Must
C3	Documents	Anyone in the loop can upload (Excel / PDF / Word). Deletion needs admin approval. Files are versioned and never hard-deleted without approval.	Must
C4	Deadline alerts	Email + WhatsApp at 50%, 85% and 100% of a step's time (configurable). Each alert goes out once and is logged.	Must
C5	Approvals + audit log	Approvals for file deletes, regularization and new staff. Append-only audit log: who did what, and when.	Must
C6	Insights & Analytics	Superset dashboards: order board heatmap, step timing, leads by category (pie), data by source, team performance.	Must
C7	Website content admin	Manage offers ticker, best sellers, categories, products, Koko knowledge and guardrails.	Must

3 Architecture and stack

One repository, a modular monolith, and one Docker Compose file that runs everything. Every box below is a container on the company server.

Container	Role	Notes
nginx	Serves the React build, proxies /api to FastAPI	Cloudflare Tunnel connects to it, so no public ports are opened.
web (React)	Public site, employee portal, admin console	One app, with routes and layouts per role. shadcn/ui + Tailwind. react-i18next.
api (FastAPI)	REST API, auth, business rules	SQLAlchemy 2 + Alembic, Pydantic v2, structlog JSON logs with request IDs.
worker	Runs queued jobs	Batch assignment, alert sending, Koko re-index, file scanning, report exports.
scheduler	APScheduler: puts jobs on the queue at set times	Batch auto-assign check (every 15 min in working hours, Mon-Sat), alert check (every 15 min), attendance auto-close (nightly), cleanup.
postgres	All data + pgvector for Koko	Separate read-only role + analytics schema for Superset.
superset	Insights & Analytics	Embedded in the admin console using guest tokens. Has its own metadata DB.

Container	Role	Notes
files volume	Uploaded documents	Stored on disk under /srv with random names. Backed up nightly with the DB (restic to B2/R2).

External services

Service	Used for	Set up in
Anthropic API	Koko answers (Haiku-class model), chat summaries	Week 0 (company key, spend limit USD 20-50/month)
WhatsApp Business Platform	Deadline alerts to employees	Week 0: start Meta Business verification now, because it can take 1-2 weeks
Resend / Brevo	Email alerts, approvals, password reset, enquiry notifications	Week 1 (domain DNS records: SPF, DKIM)
Cloudflare	DNS, HTTPS, Tunnel, Access for staging	Week 1
Sentry + UptimeRobot	Error tracking, uptime alerts	Week 7

4 Data model

Main tables, grouped by module. Every table has `id, created_at, updated_at`. Times are stored in UTC and shown in IST.

Module	Tables (key columns)
Users & access	users(name, email, phone, role[admin|employee], status[pending|active|disabled], is_available, shift_id) · sessions · availability_log(user_id, value, changed_at)
Website	offers(text, link, starts_at, ends_at, sort, active) · categories(name, slug, image, sort) · products(name, category_id, pack_format, is_best_seller, best_seller_rank, attributes JSONB) · visitor_leads(name, phone_e164, email, category_id, consent_at, source, ip_hash, assigned_to)
Koko	kb_documents · kb_chunks(embedding vector(384)) · chat_sessions(visitor_lead_id, lang) · chat_messages · koko_settings(guardrails, monthly_cap)
Attendance	shifts(name, start, end, grace_min) · attendance_sessions(user_id, check_in_at, check_out_at, note, ip, source[web|auto_closed|regularized]) · attendance_days(user_id, date, status, worked_min) · regularization_requests · holidays(date, name)
Tasks	tasks(title, category_id, source[auto_batch|order_board|admin|self|lead], batch_id, ref_type, ref_id, assigned_to, created_by, priority, due_at, status[pending|in_progress|query_open|completed], outcome[successful|lost|null], closed_at) · task_notes(task_id, author_id, body, created_at), append-only
Auto-assign	auto_assign_settings(enabled, mode[weekly|monthly|until_off|custom], start_date, end_date, work_hours, work_days, check_every_min, batch_size, max_batches_per_day, record_order) · employee_categories(user_id, category_id) · employee_assign_prefs(user_id, batch_size_override, included) · assignment_batches(id, user_id, category_id, size, status[open|completed], assigned_at, completed_at) · batch_records(batch_id, record_id, status[pending|done|invalid]) · auto_assign_checks(ran_at, stats JSONB)
Order Boards	order_boards(code UNIQUE "ABC117-xxxx", name, client, priority, start_at, deadline_at, split_mode[percent|days], status, final_status) · board_steps(board_id, seq 1-5, name, percent, days, planned_start, planned_end, actual_end, status) · board_step_members(step_id, user_id, added_by, removed_at) · step_task_templates(name, version, step_seq, category_id, client_id, active) · step_template_items(template_id, title, sort, required, needs_file, needs_approval, default_role, due_offset_days) · board_step_tasks(step_id, title, sort, required, needs_file, assigned_to, status, completed_by, completed_at, from_template_item_id) · board_documents(board_id, step_id, filename, path, sha256, size, uploaded_by, deleted_at) · delete_requests(document_id, reason, status, decided_by)
Alerts	alert_rules(threshold_pct, channels[mail,whatsapp], active) · alert_log(step_id, threshold_pct, channel, recipient, sent_at, provider_id) UNIQUE(step_id, threshold_pct, channel, recipient)

Module	Tables (key columns)
Audit	audit_log(at, actor_id or "system", action, entity_type, entity_id, before JSONB, after JSONB, ip). Append-only: a DB trigger blocks UPDATE and DELETE.

Category column. Products, leads and buyer records each carry a category (Coffee, Tea, Confectionery...). Auto-assign uses this column to route each record only to employees mapped to that category in employee_categories. Records keep an assigned_to / batch_id so a record is never in two open batches.

5 Module-by-module design

5.1 Public website

Header, offers ticker, language, Employee Login

- Ticker: a CSS animation (translateX from -50% to 0) over a duplicated list, so offers move **left to right** with no gap. It pauses on hover. Respect *prefers-reduced-motion* (show a static list instead).
- GET /api/public/offers returns only active offers within their start/end dates. Admin CRUD lives under Website content.
- Logo: KJS INDIA.
- Language: the switcher stores the choice in a cookie and the URL prefix (/en, /hi). All text goes through t() from Day 1.
- Employee Login: goes to /employee/login, which is a separate layout from the public site.

Best sellers + categories

- Best sellers: products where is_best_seller = true, ordered by best_seller_rank. Admin pins them. (Later: auto-rank by enquiries in the last 30 days.)
- Category gate: clicking a category opens a modal. Validation: name is required, plus a valid phone (E.164, using the phonenumbers library) OR a valid email. A consent checkbox is required (DPDP Act 2023).
- POST /api/public/leads creates visitor_leads and returns a signed, httpOnly cookie (visitor_id, 1 year). Categories check this cookie, so returning visitors skip the form.
- Abuse protection: rate limit per IP (e.g. 5 per hour), honeypot field, and Cloudflare Turnstile if spam appears.
- Every lead becomes a task through auto-assign (category-matched) or goes to the Unassigned queue.

Koko (AI assistant)

- Knowledge: products, categories, FAQs and policies are chunked and embedded (multilingual-e5-small, 384 dimensions) into pgvector. They are re-indexed by a job whenever content changes.
- Answer flow: retrieve the top chunks, build a system prompt ("You are Koko... answer only from context, reply in {lang}, never invent prices or capabilities"), then stream the answer.
- Lead capture: on buying intent, Koko offers "Send enquiry". It uses the gate details if known, otherwise asks for name + phone/email, then creates a task with the chat summary.
- Controls: 20 messages per session, 400 tokens per reply, a monthly spend cap (chat falls back to a contact form when the cap is reached), and IPs stored as hashes.

5.2 Attendance

- POST /attendance/check-in and /check-out use server time only. A partial unique index allows one open session per user.
- Multiple sessions per day are summed. A night shift is credited to its start date.
- Nightly job: closes open sessions at shift end (source = auto_closed), computes attendance_days (Present / Absent / Weekly off on Sunday / Holiday / Leave / Late), and asks the employee to regularize.
- Regularization: the employee requests times + a reason, the admin approves or rejects, and it is written to audit_log.

- Optional: record the IP on check-in and flag check-ins from outside the office network. Location tracking is not planned (privacy).

5.3 Availability toggle

- PATCH /me/availability {is_available}. Each change is written to availability_log and audit_log.
- Auto-assign only sends a batch to employees who are active, available, have no open batch, and are not on leave or absent that day. Optional rule (admin setting): must also be checked in.
- Turning availability OFF does not take back a batch already assigned; the employee finishes it or the admin reassigns it. No new batch arrives while OFF.

5.4 Tasks and dashboard stats

Stat	Definition (per employee; admin sees the sum over all employees)
Total tasks	Tasks assigned in the selected period
Completed	status = completed
Pending	status in (pending, in_progress), with overdue shown separately (due_at < now)
Ongoing client queries	status = query_open (a client conversation is in progress)
Successfully closed	outcome = successful (closed_at within the period)

These are computed by one SQL view (task_stats_v) grouped by user_id and period, so the employee and admin dashboards always agree. A record batch counts as one task; its progress (e.g. 12 of 30) comes from batch_records.

- **Add my own task** (dashboard and My tasks): title, category, due date, priority, optional link to a client or board, and notes. source = self. It counts in the employee's stats like any other task. The admin can see these tasks too.
- **Notes**: POST /tasks/{id}/notes. Notes are append-only (a note can be corrected only by adding a new one), have the author and time, and are visible to the employee and admin. Order board tasks use the same notes table.

5.5 Auto task assign (record batches)

Tasks for auto-assign are **records from our database** (buyer records, leads, imported rows). Each record has a category. Instead of a fixed daily run, the system hands out **batches**: when an employee is free, they get the next N records of their categories. N is set by the admin in the box "Records per batch (max per employee)".

Settings

- **enabled** toggle. When switched ON, the admin picks: **Weekly**, **Monthly**, **Until I turn off**, or **Custom** (start/end dates). When the window ends, auto-assign switches itself off and logs it.
- **work_days** Mon-Sat (Sunday off) + holidays, and **work_hours** (e.g. 09:30-18:30). No batches are sent outside these.
- **batch_size**: quick buttons 20 / 30 / 50 or any number (allowed range 1-500). Optional **per-employee override** (e.g. Karan 50, Meena 20).
- **Send the next batch when**: the current batch is completed (default). Optional: when it is X% done.
- **max_batches_per_day** per person (e.g. 3) as a safety limit, and **record_order** (oldest first / newest first / priority).
- **employee_categories**: many-to-many, edited by the admin. Example: Priya = Tea, so only Tea records go to Priya.

How a batch is assigned (job runs every 15 minutes, plus on events)

```
if not enabled or now outside work_days/work_hours or window ended: stop
free = employees WHERE active AND is_available AND included
         AND NOT EXISTS (open batch for this employee)
         AND batches_today(employee) < max_batches_per_day
         [AND checked_in today  -- optional rule]
for emp in free (longest-waiting first):
    size = emp.batch_size_override or settings.batch_size
    BEGIN
      rows = SELECT id FROM records
             WHERE category_id IN emp.categories AND assigned_to IS NULL
             ORDER BY created_at
             LIMIT size
             FOR UPDATE SKIP LOCKED      -- two jobs can never grab the same rows
      if rows is empty: continue         -- nothing waiting in their categories
      batch = INSERT assignment_batches(user=emp, size=len(rows), status='open')
      UPDATE records SET assigned_to=emp, batch_id=batch WHERE id IN rows
      INSERT tasks(source='auto_batch', batch_id=batch)  -- 'Tea records batch #B-2210'
      audit + notify employee (in-app; email optional)
    COMMIT
categories with waiting records but no eligible employee -> admin alert (once a day)
```

- **Events also trigger it**, so nobody waits 15 minutes: an employee marks a batch completed, switches Available ON, or is given a new category.
- A batch is **completed** when all its records are marked done or invalid (or the employee clicks "Mark completed" with a note). Then the next batch can go out.
- If an employee goes unavailable or on leave with an open batch, the admin can **reassign** or **release** it (records go back to the pool). Optional rule: auto-release after X hours with no activity.
- **Preview** runs the same code without saving ("Priya free, gets 30 Tea records; Rahul busy, 18 left"). **Run now** queues the same job.
- **Records waiting** per category = COUNT(*) WHERE assigned_to IS NULL, shown on the admin screen so gaps (no one handles Seasonings) are visible.

5.6 Order Boards

Creating a board

- "New order board" modal: name, ID suffix (the prefix ABC117- is fixed). Validated with the regex `^ABC117-[0-9]{1,6}$` and a UNIQUE constraint. Also: client, priority, final deadline, requirement files (Excel / PDF / Word, max 25 MB each), and the step 1 owner.
- The 5 steps are created from a template: 1 Procurement, 2 Packaging, 3 Production, 4 Placement, 5 Track (final status: Delivered / Returned / Cancelled).
- A board is like a folder: files, steps, members, alerts and history all belong to it. GET /order-boards/{code} returns everything for the detail screen.

Time split: percentage or days

```
W = working days between start_at and deadline_at (Mon-Sat, minus holidays)
percent mode:  step_days[i] = W * percent[i] / 100   (percents must total 100)
days mode:     step_days[i] = entered by admin       (total must be <= W, otherwise warn)
planned_start[1] = start_at
planned_end[i]   = add_working_days(planned_start[i], step_days[i])
planned_start[i+1] = planned_end[i]
elapsed_pct(step) = working time elapsed since step start / step working time * 100
```

- Default split (editable per board and as a global default): Procurement 25%, Packaging 19%, Production 30%, Placement 16%, Track 10%.
- When a step finishes early or late, the next step starts at the actual finish time. The admin chooses: **keep the final deadline** (remaining steps are re-scaled) or **keep step durations** (the final date moves and the board is flagged).

- Status colours: On track (< 50%), Watch (50-85%), At risk (85-100%), Overdue (> 100%).

Members, documents, approvals

- Members are per step. The admin and **any current member** can add a colleague ("+ Add employee" with search). Only the admin can remove someone (soft removal with removed_at). Usually one person is on step 1 and more join later. Employees only see boards where they are a current member.
- Upload: any member or admin. Files are stored with a random name, sha256 and MIME/size checks. There is a version history (re-uploading the same name creates v2).
- Delete: a member creates a delete_request, and the file stays visible with a "Delete requested" badge. The admin approves (soft delete: deleted_at set, file kept for 30 days) or rejects. Admin deletes are also logged.
- Completing a step: allowed only when all **required** step tasks are completed. The step owner clicks "Mark step complete". This sets actual_end, starts the next step and notifies its members by email and WhatsApp.

Step tasks: how they are fed in (recommended approach)

The tasks inside each step differ from client to client, so they should be **data, not code**. Recommended design, in three layers:

Layer	How it works	When
1. Step task templates	Admin screen "Step templates": for each of the 5 steps, a named checklist (e.g. "Coffee premix - Standard v2"). Each item has: title, order, required yes/no, needs a file, needs approval, default role/person, due offset (e.g. day 2 of the step). Templates can be linked to a **category** or a **specific client**, and are versioned.	Build in Week 5
2. Copy into the board	When a board is created, the matching template is picked automatically (client template, else category template, else the default), and the admin can change it. Items are **copied** into board_step_tasks, so later template edits never change running boards.	Week 5
3. Edit per board	Admin can add, remove, reorder and reassign tasks on a live board. Members can add extra tasks. Every change is in the audit log.	Week 5
Later: AI suggestion	Claude reads the uploaded requirement file (Excel/PDF/Word) and **suggests** step tasks (e.g. "Order 2 g stick film", "Allergen check: milk"). The admin accepts or edits each one before it is added. Nothing is added without a human.	Release 2
Also possible	Import templates from an Excel sheet (step, task, required, needs_file), so the client can maintain them without the admin UI.	Release 2

This keeps the code identical for every client: only the template data changes. The employee screen shows "Tasks completed 3 of 5", with who completed each task and when.

5.7 Deadline alerts (email + WhatsApp)

- The scheduler runs **check_deadlines** every 15 minutes during working hours. For each active step it computes elapsed_pct. For each threshold (50, 85, 100 by default) that has been crossed and is not yet in alert_log, it queues the send.
- The UNIQUE key on alert_log means each alert goes out **once**, even if the job runs twice. Failures retry 3 times with backoff and are logged.
- Recipients: current step members + admin (configurable). Optional repeat: daily reminders after 100% until the step is done.
- WhatsApp: messages to staff outside a 24-hour chat window must use **Meta-approved templates**. Submit 3 templates in Week 5 (50% / 85% / 100%), e.g. "Hi {{1}}, step {{2}} of order {{3}} has used {{4}}% of its time. Deadline: {{5}}." Employees must opt in to WhatsApp messages (collect this at onboarding).
- Email: the same 3 templates sent via Resend/Brevo, with a link to the board.

5.8 Audit log

- Written in the same database transaction as the change (no separate service), so an action can never happen without its log row.
- Covers: logins, check-ins, availability, task changes, auto-assign runs and settings, board create/edit, member add/remove, uploads, delete requests and decisions, alerts sent, approvals, and every admin setting.
- A DB trigger raises an error on UPDATE or DELETE of audit_log. The admin UI has filters (user, area, date) and Excel export.

5.9 Insights & Analytics (Apache Superset)

- Run Superset as its own container. It connects with a **read-only** Postgres role that can only see an **analytics** schema of SQL views (no passwords, no raw personal data).
- Embed in the admin console using the Superset Embedded SDK: the API issues a short-lived guest token for the logged-in admin. Superset is never exposed to the public.
- Dashboards: **Order boards** (heatmap of average days late by step and week, on-time %, boards by current step, time used per step), **Leads & sources** (leads by category pie, records by source bar, gate conversion, Koko conversations to leads), **Team** (tasks done, closed successful, attendance hours, availability %).
- Views to create: analytics.board_step_timing, analytics.leads_daily, analytics.records_by_source, analytics.task_stats, analytics.attendance_daily.
- Fallback: if Superset takes too long, ship the same 4-5 charts natively in React (Recharts) first and add Superset later.

6 Roles and permissions

Action	Visitor	Employee	Admin
Browse home + best sellers, chat with Koko	Yes	Yes	Yes
Open a category	After the gate form	Yes	Yes
Check in/out, availability toggle, own tasks and stats	-	Yes	Yes
See order boards	-	Only boards they are a member of	All
Upload documents to a board	-	Yes (as a member)	Yes
Delete a document	-	Request only	Approve / delete
Add a colleague to a board step	-	Yes (as a member)	Yes
Remove board members, edit time split, manage step templates	-	-	Yes
Add own tasks, add notes to tasks	-	Yes	Yes
Auto-assign settings, batch size, category mapping	-	-	Yes
Approve staff, regularization, deletes	-	-	Yes
Audit log, Insights & Analytics, website content	-	-	Yes

Every API endpoint has a role dependency and a test for both the allowed and the forbidden case (rule from CLAUDE.md).

7 Week-by-week plan

Mon-Fri development days. Each week ends with a deploy to staging and a short demo. Work in small, tested, committed steps with Claude Code (plan mode first, then implement, test and commit).

Wk	Dates	Build	Done when
0	Wed 14 - Fri 16 Oct	Freeze requirements (this plan + wireframes v2). Tools, repo, CLAUDE.md. Start **Meta Business verification** for WhatsApp. Ask IT about the server. Send the inputs list to the client.	Client signs off wireframes; WhatsApp verification submitted.
1	Mon 19 - Fri 23 Oct	Skeleton (FastAPI, React, Postgres, Docker), logging, CI. Auth: employee sign-up, admin approval, login, roles. audit_log + trigger. Working calendar + holidays. Staging deploy via Cloudflare Tunnel.	https://staging... works; login + approval flow tested.
2	Mon 26 - Fri 30 Oct	Public site: header, offers ticker, logo link, language switch (EN + HI), best sellers, categories, category gate + visitor_leads + cookie. Admin: offers, products, categories CRUD.	A visitor can unlock a category; the lead appears in admin.
3	Mon 2 - Fri 6 Nov	Employee portal: attendance (check-in/out, weekly view, auto-close, regularization), availability toggle, tasks + statuses, **own tasks**, **task notes**, dashboard stats (task_stats_v). Admin: team & attendance, approvals.	Employee works a full day in staging; admin sees correct totals.
4	Mon 9 - Fri 13 Nov	Auto task assign: settings UI, batch size (+ per-employee override), category mapping, batch job (SKIP LOCKED) + event triggers, preview, run now, batch history, records-waiting counts. Koko v1: knowledge index, chat widget, guardrails, lead capture.	A free, available employee gets N records of their category; the next batch arrives after completion. Koko creates leads. *(Optional soft launch of the public site.)*
5	Mon 16 - Fri 20 Nov	Order Boards core: create (ABC117 ID), files, 5 steps, % / days split, members (employees can add colleagues), **step task templates** + per-board checklists, step completion and handoff, history, employee view. Submit WhatsApp templates.	A board moves through all 5 steps with correct dates.
6	Mon 23 - Fri 27 Nov	Document delete approvals + versioning. Deadline alerts: check_deadlines job, alert_log, email templates, WhatsApp sending, alert settings UI.	50/85/100% alerts arrive by mail + WhatsApp exactly once.
7	Mon 30 Nov - Fri 4 Dec	Superset: container, read-only role, analytics views, 3 dashboards, embedded in admin. Hardening: security checklist, backups + restore drill, Sentry, uptime, Playwright E2E.	Dashboards load inside admin; E2E green; restore drill passes.
8	Mon 7 - Fri 11 Dec	UAT with the client (scripts for visitor, employee, admin). Fix P1/P2 bugs. Production deploy, seed real data, user guides. Go-live Fri 11 Dec.	Live on the production domain; client sign-off.

Effort by module (one developer + Claude Code)

Module	Days
Foundation, auth, roles, audit log, deploy	5
Public site (ticker, best sellers, categories, gate, i18n) + content admin	4.5
Koko (RAG, widget, guardrails, lead capture)	3
Attendance + regularization	3
Availability, tasks, own tasks, notes, dashboard stats	2.5
Auto task assign	3
Order Boards core + step task templates	5.5

Module	Days
Documents + delete approvals	1.5
Deadline alerts (mail + WhatsApp)	3
Superset analytics	3
Hardening, tests, backups, monitoring	3
UAT, fixes, go-live	5
Total	**~42 working days**

8 Testing, go-live and risks

Testing

- Unit tests for the tricky logic: working-day maths (Sundays, holidays), time split, elapsed %, batch assignment (no record in two batches, no batch to a busy or unavailable employee, batch size override), template copy, alert thresholds, ID validation.
- API tests for every endpoint: allowed + forbidden role, validation errors.
- Playwright E2E: visitor unlocks a category -> lead auto-assigned; employee check-in -> task -> close as successful; admin creates board -> steps -> 85% alert -> delete approval.
- Clock control in tests (freezegun) so deadline and alert tests are deterministic.

Risks and fallbacks

Risk	Impact	Fallback
WhatsApp Business verification or template approval is slow	Alerts delayed	Start in Week 0. Ship email alerts first; WhatsApp switches on when approved. A BSP (Gupshup/Interakt) can speed up onboarding.
Server not ready in Week 1	Deploys blocked	Use a small cloud VM with the same Docker Compose; move later (about half a day).
Superset embedding takes longer	Analytics late	Native React charts for the 4-5 key charts first, Superset next.
Scope creep (new requests)	Go-live slips	docs/BACKLOG.md with date and requester. "Planned for Release 2, unless we swap it for X."
Category data missing on records	Auto-assign cannot route	Make category required on import; uncategorised items go to the Unassigned queue.
Spam leads via the category gate	Bad data	Rate limit, honeypot, Turnstile; optional OTP in Release 2.
Koko gives a wrong answer	Trust	Answer only from retrieved context, never quote prices, "I will connect you with sales" fallback, conversation review in admin.

9 Inputs needed from the client (send on Thu 15 Oct)

Input	Needed by
Sign-off on wireframes v2 and this plan (including the 11 Dec go-live)	Fri 16 Oct
Meta Business account details + WhatsApp number for alerts	Fri 16 Oct
Product list with category, pack format and best sellers; category images	Fri 23 Oct
Languages for launch (proposed: English + Hindi)	Fri 23 Oct
Logo, brand colours, banner photos, first offers for the ticker	Fri 23 Oct
Employee list: name, email, phone, shift, categories; holiday list 2026-27	Fri 30 Oct
FAQs and policies for Koko	Fri 6 Nov
Default time split for the 5 steps; who receives alerts	Fri 13 Nov
2-3 real requirement files (Excel/PDF/Word) to test Order Boards	Fri 13 Nov

Open questions

- Should auto-assign require the employee to be **checked in**, or only **available**?
- Default batch size (20, 30 or 50)? Should an unfinished batch be released back to the pool after some hours?
- Which step task templates does the client want at launch (one standard set, or per client/product)?
- Do leads from the category gate need OTP verification at launch, or is validation enough?
- When a step runs late, should the final deadline stay fixed (compress later steps) or move?
- Does the client want attendance data exported in a specific payroll format?
- Should employees receive WhatsApp alerts on their personal numbers (needs opt-in), or only email?